

PROJECT REPORT

Title: Auto Ticket Classification using Flow Designer

Team Id: SWTID-2026-4440

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1. Introduction

1.1 Project Overview

Auto Ticket Classification using Flow Designer is a no-code automation solution built on the ServiceNow platform for a school IT helpdesk. The helpdesk receives a large number of incident requests every day from students and teachers covering issues such as Wi-Fi connectivity, projector failures, password/login problems, and slow computers. Traditionally, IT staff had to manually read every incoming request, decide which category and subcategory it belonged to, and route it to the right team. This manual triage step was time-consuming, inconsistent, and did not scale as the volume of tickets increased.

The project automates this classification step using ServiceNow Flow Designer. When a new incident is created, a flow automatically scans the keywords present in the Short Description field and, based on predefined keyword rules, assigns the correct Category and Subcategory to the ticket without any manual intervention. Once classified, the system also sends an automatic email acknowledgement to the caller confirming that their request has been logged.

1.2 Purpose

The purpose of this project is to reduce the manual workload on IT support staff, eliminate classification errors caused by human oversight, and speed up ticket routing so that issues reach the right resolver group faster. By using a no-code Flow Designer approach instead of custom scripting, the solution is also easy for non-developers to understand, maintain, and extend over time — making it well suited to a small IT team with limited technical resources.

2. Ideation Phase

2.1 Problem Statement

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers. Currently, IT staff manually reviews each request and assigns a category, which is time-consuming and inefficient. There is a need for a system that can automatically classify incoming tickets at the time of creation, based on the wording of the request itself, so that staff effort is spent resolving issues rather than sorting them.

2.2 Empathy Map Canvas

An empathy map was used to understand the experience of the two key user groups affected by the manual ticket-classification process: the students/teachers who raise tickets, and the IT support staff who process them.

IT Support Staff

Aspect	Observation
Says	"I spend too much time reading every ticket just to figure out where it belongs."
Thinks	Worries that repetitive manual sorting is not the best use of their time and that mistakes will slow down resolution.
Does	Manually opens each incident, reads the description, and selects a Category/Subcategory before assigning it.
Feels	Frustrated by repetitive work and pressured when ticket volume spikes during exam or admission periods.

Students / Teachers (Callers)

Aspect	Observation
Says	"I just want my issue logged quickly and to know someone is looking into it."
Thinks	Expects a fast acknowledgement and doesn't want to fill in technical fields like Category or Subcategory.
Does	Submits a ticket with only a caller name and a short description of the problem.
Feels	Reassured when they receive an immediate confirmation email; frustrated by long response times.

2.3 Brainstorming

The team brainstormed several possible approaches to solving the classification problem before settling on a Flow Designer based, keyword-driven solution:

- Build a custom table (Incident Workflow) to capture ticket data in a structured, standardized format.
- Use keyword matching on the Short Description field (e.g. "Wi-Fi", "Projector", "Password", "Slow") to infer the correct Category and Subcategory.
- Implement dependent choice lists so Subcategory options automatically narrow down based on the selected Category, keeping the data consistent.
- Use a no-code Flow Designer flow (Trigger → Flow Logic → Update Record → Send Email) instead of server-side scripting, so the logic stays easy to read and maintain.
- Automatically notify the caller by email once their ticket is classified and logged, closing the feedback loop.

Of these ideas, the keyword-based Flow Designer automation was selected because it required no scripting knowledge, could be built and modified directly in the ServiceNow UI, and could be demonstrated end-to-end within the scope of the project.

3. Requirement Analysis

3.1 Customer Journey Map

Stage	Caller Action	System Response
1. Raise Ticket	Enters Caller Name and a Short Description of the issue only.	A new record is created in the Incident Workflow table with Category left empty.
2. Automatic Classification	No action required.	Flow Designer trigger fires on record creation and scans the Short Description for keywords.
3. Category Assignment	No action required.	Flow Logic evaluates the keyword and updates Category and Subcategory automatically.
4. Notification	Receives an email.	An automated email is sent to the caller confirming the ticket has been submitted.
5. Resolution Routing	Waits for support.	The correctly categorized ticket is available for the right IT team to pick up and resolve.

3.2 Solution Requirement

Functional Requirements

- Automatically classify IT tickets based on the issue description.
- Assign both Category and Subcategory without manual intervention.
- Support dependent choice logic between Category and Subcategory.
- Send an automated email notification to the caller upon ticket creation.
- Store ticket information in a structured and standardized format.

Non-Functional Requirements

- Usability – no-code configuration that non-developers can understand and maintain.
- Scalability – must continue to work as ticket volume grows, without redesign.
- Maintainability – easy to add new keyword rules or categories in the future.
- Reliability – consistent, accurate classification and guaranteed email delivery.
- Security – only authorized users can access or modify ticket data.

3.3 Data Flow Diagram

The diagram below describes the logical flow of data through the system, from ticket creation to caller notification:

- Caller (Student/Teacher) → submits Caller Name + Short Description → Incident Workflow table (new record, Category empty).
- Incident Workflow table (Record Created, Category is Empty) → triggers → “Auto Classify School IT Tickets” Flow.
- Flow Logic → evaluates Short Description keywords (Wi-Fi / Network, Projector, Password / Login, Slow / Hanging).
- Matched condition → Update Record action → sets Category + Subcategory on the Incident Workflow record.
- Update Record action → Send Email action → email delivered to Caller confirming ticket submission.

3.4 Technology Stack

Layer	Technology / Component
Platform	ServiceNow
Automation Engine	ServiceNow Flow Designer (no-code)
Data Storage	Custom table – Incident Workflow
Field Types Used	Auto Number, Reference (sys_user, sys_user_group), Choice, String
Notification	ServiceNow Send Email action
Change Management	ServiceNow Update Sets (for migrating configuration)

4. Project Design

4.1 Problem – Solution Fit

Problem	Solution
Manual reading of every ticket to decide its category is slow and inconsistent.	Flow Designer automatically evaluates the Short Description text as soon as the ticket is created.
Subcategory options are not related to the chosen Category, risking invalid combinations.	A dependent choice field restricts Subcategory options based on the selected Category.
Callers do not receive confirmation that their request was received.	An automated email is sent to the caller immediately after classification.
A scripted solution would be hard for non-technical IT staff to maintain.	The entire logic is built with no-code Flow Designer actions and conditions.

4.2 Proposed Solution

The proposed solution introduces a dedicated Incident Workflow table that captures each ticket with fields for Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, and Assigned To. When a caller submits only a Caller Name and Short Description, a Record Created trigger (condition: Category is empty) starts a Flow Designer flow. The flow uses a series of If / Else If branches to check the Short Description for the keywords Wi-Fi/Network, Projector, Password/Login, and Slow/Hanging. Whichever branch matches, an Update Record action sets the corresponding Category and Subcategory, and a Send Email action notifies the caller. This removes the need for any manual triage step by IT staff.

4.3 Solution Architecture

At a high level, the solution follows a simple trigger → logic → action architecture native to ServiceNow Flow Designer:

- Data Layer: Incident Workflow custom table stores all ticket fields, choices, and dependencies.
- Automation Layer: Flow Designer flow (“Auto Classify School IT Tickets”) with a Record Created trigger and Flow Logic branches.
- Action Layer: Update Record actions write Category/Subcategory; a Send Email action notifies the caller.
- Governance Layer: An Update Set (“Project Update Set”) tracks all configuration changes and can be exported/migrated to other instances.

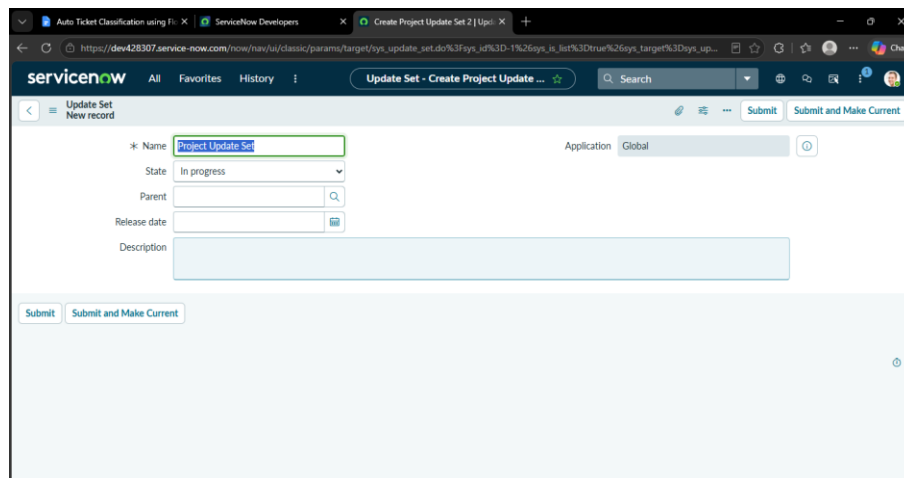
5. Project Planning & Scheduling

5.1 Project Planning

The project was executed in six phases, moving from requirement gathering through to deployment and documentation:

Phase	Milestone / Activities	Key Outcome
Phase 1	Requirement Analysis & Planning – business use case, creation of Update Set	Business requirements finalized; “Project Update Set” created and set as current
Phase 2	Backend Development & Configuration – custom table, field creation, choice dependency	Incident Workflow table built with all required fields and a working Category→Subcategory dependency
Phase 3	Automation using Flow Designer & Email Notification	Flow built with trigger, keyword-based logic branches, and email action
Phase 4 / 6	Testing, Validation & Security	Two end-to-end test scenarios executed and validated
Phase 5	Deployment & Final Documentation	Update Set marked Complete, exported to XML, and project documented

5.2 Project execution process output



The screenshot shows the ServiceNow interface for creating a new Project Update Set. The form includes the following fields and controls:

- Name:** A text field containing "Project Update Set".
- State:** A dropdown menu currently set to "In progress".
- Application:** A field set to "Global".
- Parent:** A search field with a magnifying glass icon.
- Release date:** A date selection field with a calendar icon.
- Description:** A large, empty text area for additional details.
- Buttons:** "Submit" and "Submit and Make Current" are located at the bottom left of the form.

Fig 5.2 – Creating the Project Update Set

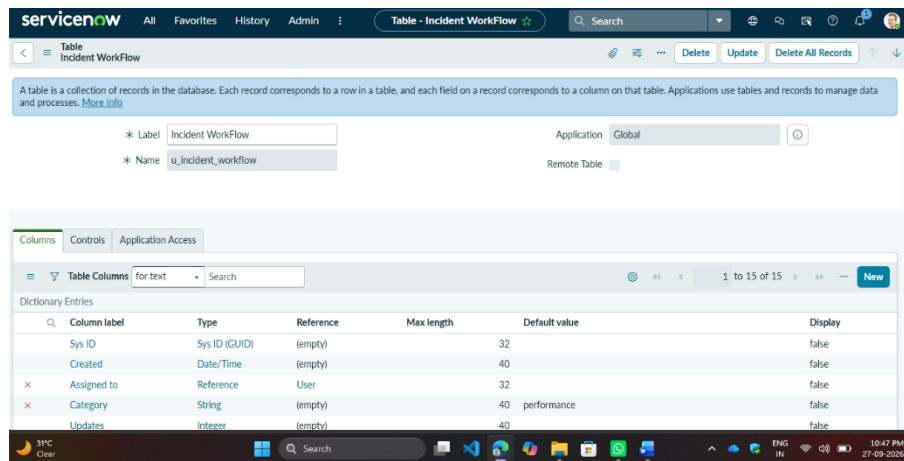


Fig 5.2 – Creation of Custom Table to Store Ticket Records

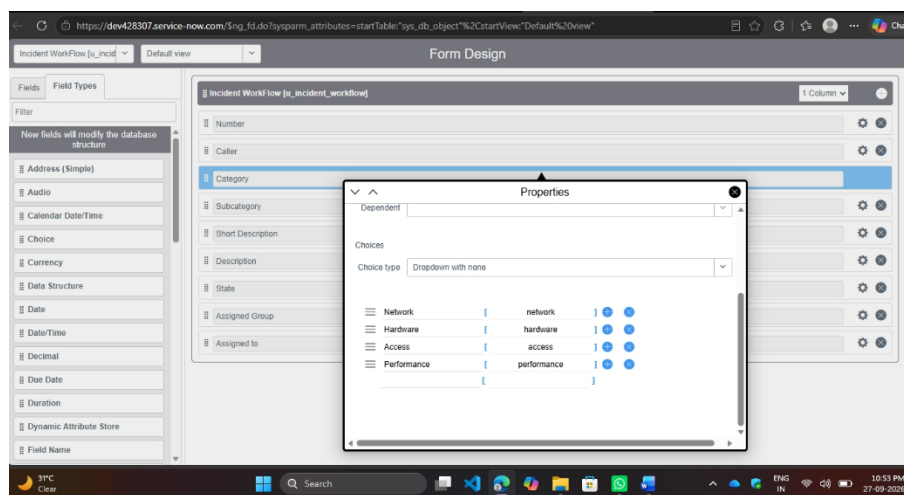


Fig 5.3 – Field Creation and Data Type Configuration

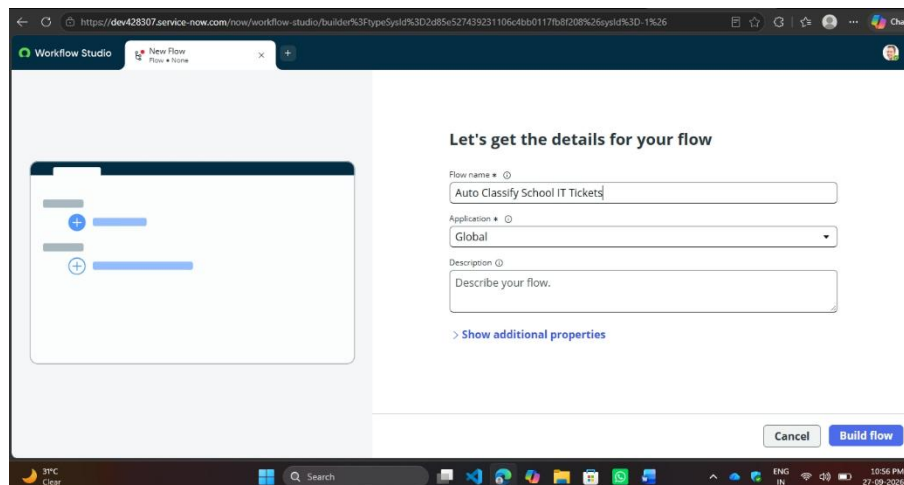


Fig 5.4 – Auto Classification using Flow design

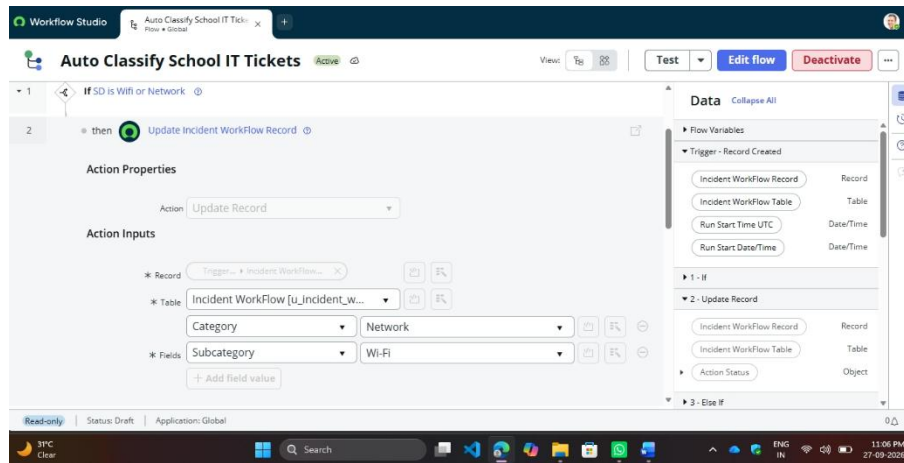


Fig 5.5 – Trigger Configuration

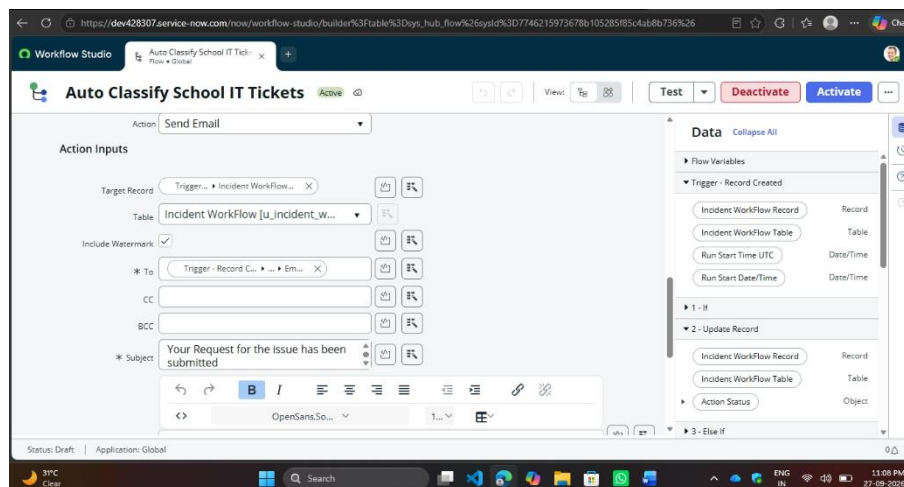
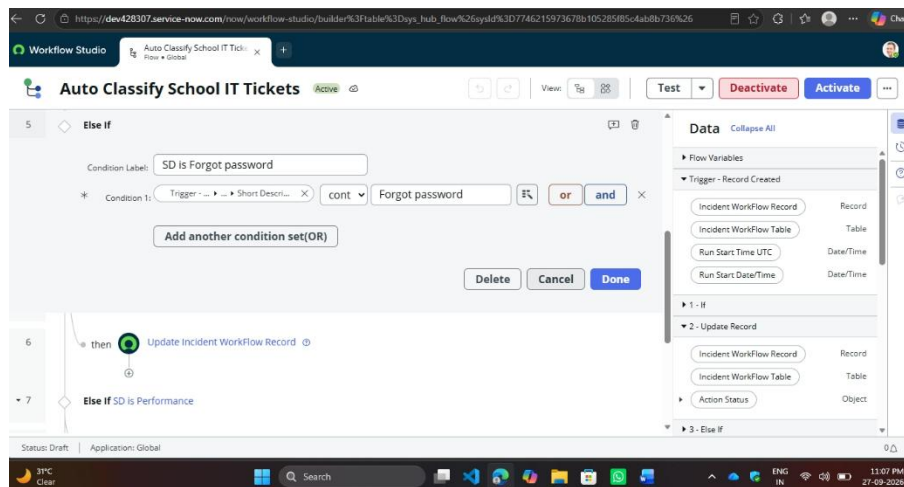


Fig 5.6 – Email Action Setup

6. Functional and Performance Testing

6.1 Performance Testing

The following data validation checks were carried out to confirm the solution performs correctly:

- Mandatory fields captured correctly.
- Auto-number generated without duplication.
- Category and Subcategory values stored accurately.
- Reference fields resolve correctly to user and group tables.

Test Scenario 1 – Wi-Fi Issue

- Input: Caller Name entered; Short Description = “WiFi not working in library”.
- Expected Result: Category = Network, Subcategory = WiFi, confirmation email sent.
- Actual Result: Matched – fields auto-filled correctly and email delivered to the caller.

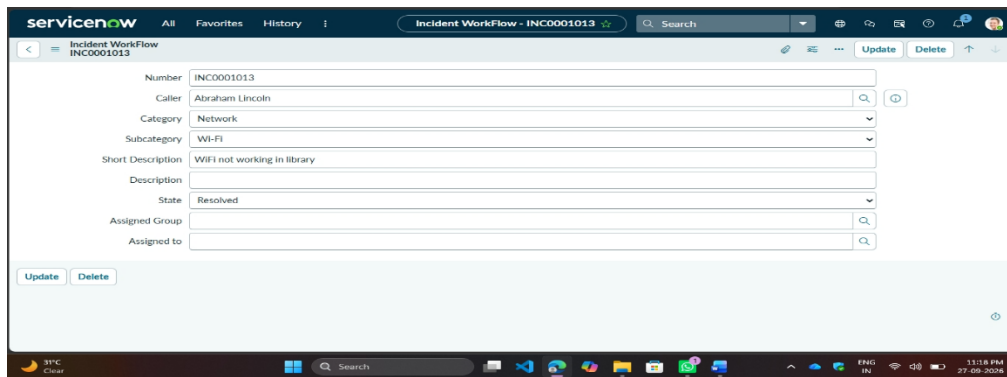


Fig 6.1 – Test Scenario 1 result: Category and Subcategory auto-filled

Test Scenario 2 – Projector Issue

- Input: Caller Name entered; Short Description = “Projector not turning on.”
- Expected Result: Category = Hardware, Subcategory = Projector, confirmation email sent.
- Actual Result: Matched – fields auto-filled correctly and email delivered to the caller.

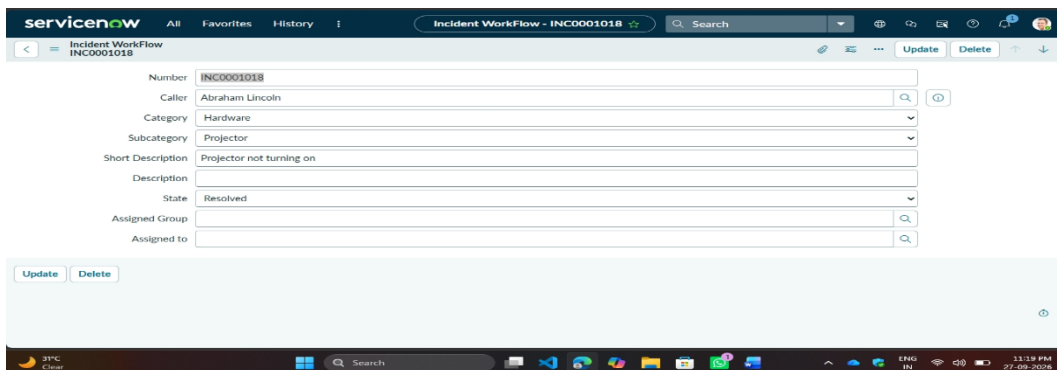


Fig 6.2 – Test Scenario 2 result: Category and Subcategory auto-filled

Both test cases confirmed that the flow reliably classifies tickets from keyword matches and that email notifications are triggered without manual effort, meeting the functional requirements defined in Section 3.2.

7. Results

7.1 Output Screenshots

Representative screenshots from the working solution are shown below. Additional configuration screenshots are available in the original project file.

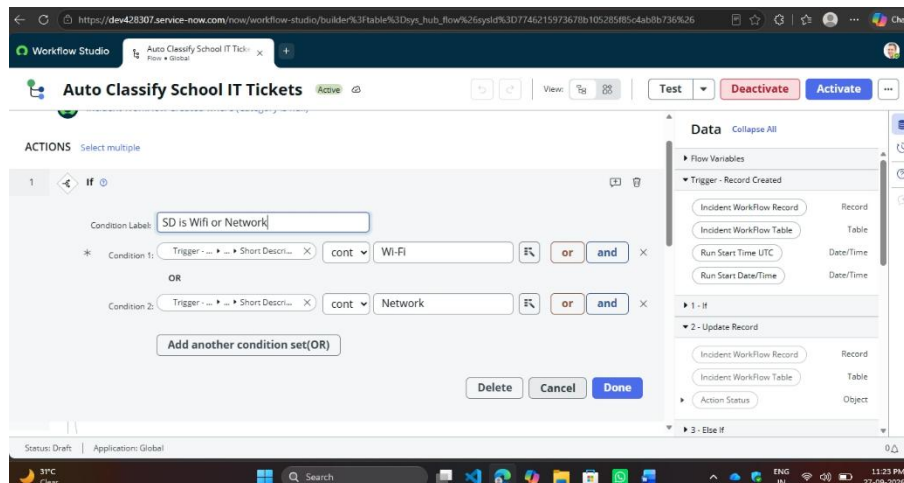
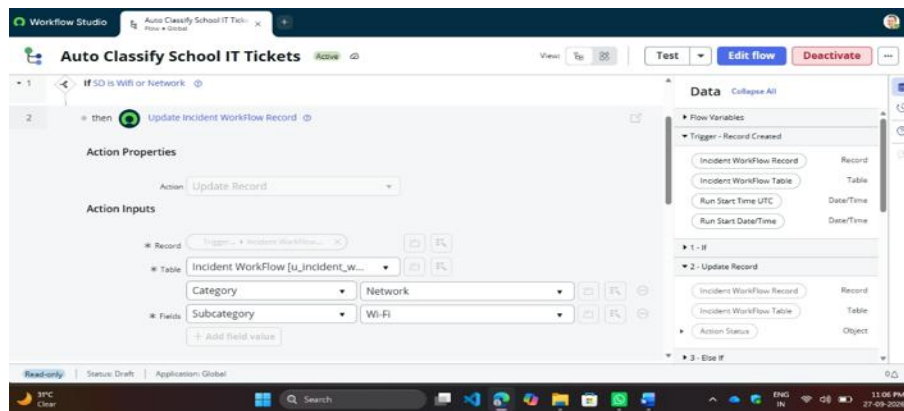


Fig 7.1 – Flow Logic condition for Wi-Fi / Network classification

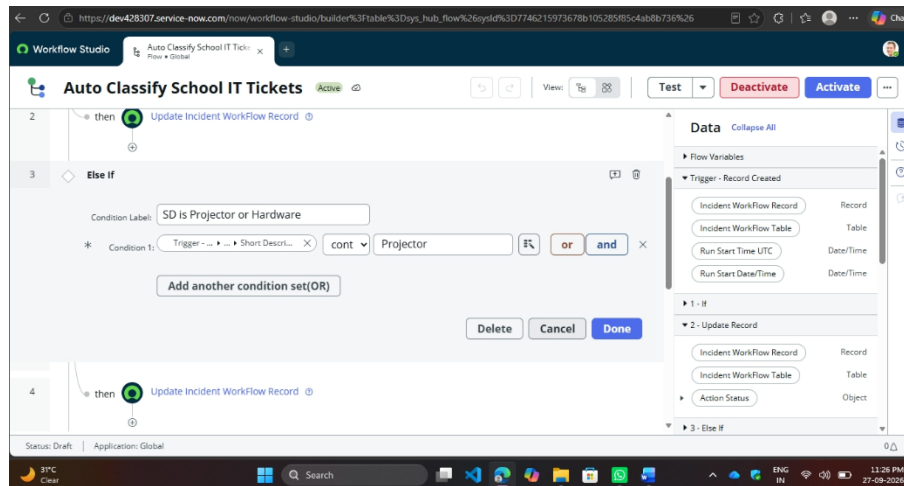


Fig 7.2 – Update Record action setting Category = Hardware, Subcategory = Projector

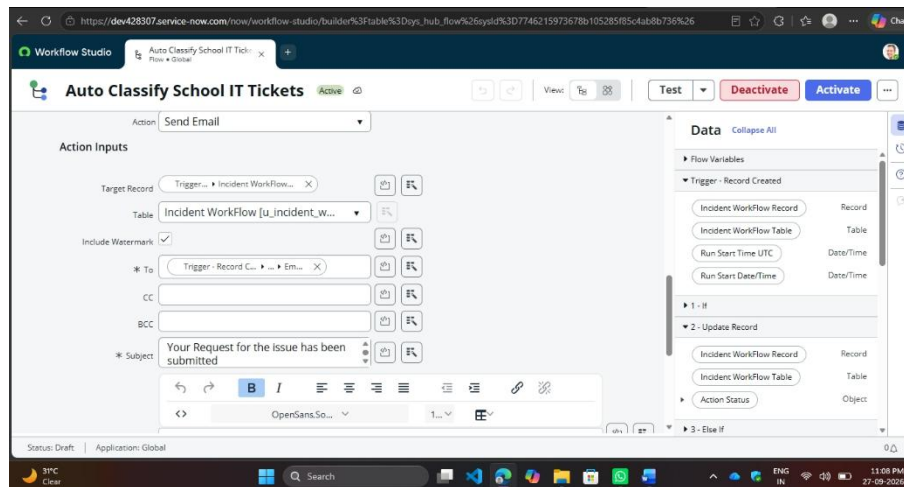


Fig 7.3 – Send Email action configuration

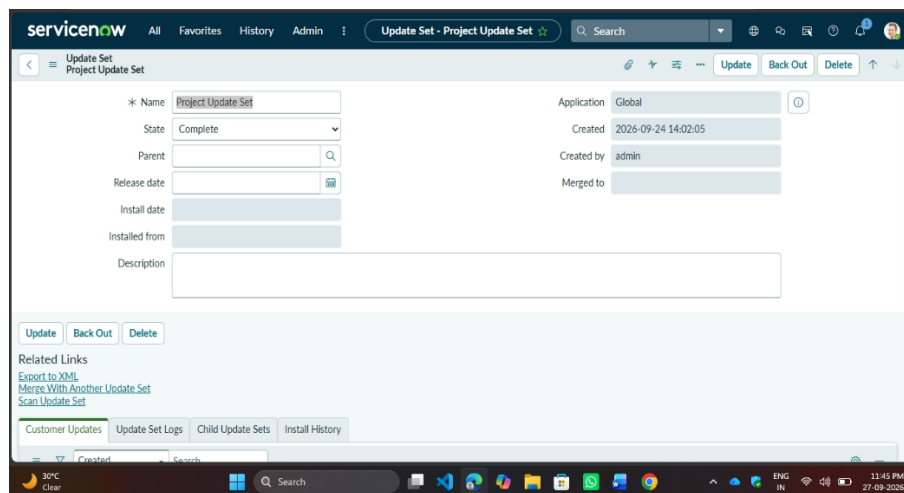


Fig 7.4 – Update Set exported to XML for deployment

8. Advantages & Disadvantages

Advantages

- Eliminates manual ticket sorting, saving IT staff time and reducing human error.
- No-code implementation makes the solution easy to understand, audit, and maintain.
- Dependent choice fields keep Category/Subcategory data clean and consistent.
- Automated email acknowledgement improves the caller experience and communication.
- Update Set packaging makes the configuration easy to move between instances.

Disadvantages

- Relies on simple keyword matching, so tickets that describe an issue without using the expected keywords may be misclassified or left unclassified.
- Adding a new issue type requires manually creating a new Flow Logic branch and choice values, which does not scale automatically.
- Does not use machine learning, so it cannot learn from past misclassifications or handle ambiguous, free-form phrasing.
- Limited to the Short Description field; it does not consider the longer Description field or attachments.

9. Conclusion

The Auto Ticket Classification using Flow Designer project successfully delivers a complete, end-to-end automation solution for a school IT helpdesk. The primary objective of reducing manual effort in ticket categorization and improving response efficiency has been fully achieved.

Through structured requirement analysis, careful backend configuration, and no-code automation using Flow Designer, the system now automatically classifies IT tickets based on the short description provided by students. The implementation of dependent choice fields ensures data accuracy, while automated email notifications improve communication by instantly acknowledging ticket creation to the caller.

The solution was developed using a phased approach, ensuring clarity, stability, and ease of maintenance. Comprehensive testing and security validation confirmed that the system functions reliably, maintains data integrity, and enforces appropriate access controls.

Overall, this project demonstrates how ServiceNow can be effectively used to build scalable, maintainable, and user-friendly automation solutions without relying on complex scripting or machine learning.

10. Future Scope

- Replace or supplement keyword matching with an NLU/ML-based classifier for higher accuracy on free-form descriptions.
- Extend the keyword/category list to cover additional issue types beyond Network, Hardware, Access, and Performance.
- Add SLA tracking and automatic escalation for tickets that remain unresolved beyond a defined time.
- Introduce automated assignment of tickets to specific technicians based on category and workload.
- Build a simple caller-facing status page or portal so students/teachers can track their ticket without emailing IT.
- Add analytics/reporting dashboards to track ticket volume trends by category over time.

11. Appendix

Source Code

This solution is implemented entirely using no-code ServiceNow Flow Designer configuration (triggers, flow logic, and actions) rather than scripted code. The complete configuration is packaged in the “Project Update Set” and exported as XML (see Fig 7.4).

Dataset Link

Not applicable – the project uses live ServiceNow incident records rather than an external dataset.

GitHub & Project Demo Link

GitHub repository link: <https://github.com/Densingh-xz11/Auto-Ticket-Classification-using-Flow-Designer.git>

Project demo video link : <https://drive.google.com/file/d/1EAFir-xsNjLOKYLJhxNjbsdJkv23nNqe/view?usp=drivesdk>